HG-MARL Based Scheduling of Trajectory and Offloading for Layered UAVs-Enabled Digital Twin Channel Modeling in Integrated Communication-Computing-Intelligence-Controlling Network

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***Abstract*—As a key paradigm in sixth-generation (6G) systems for enabling physical–virtual mapping and intelligent network management, Digital Twin (DT) networks demand high-precision, low-latency modeling and continuous updates of wireless environments. However, in dynamic urban scenarios, channel conditions are highly non-stationary, and traditional sensing schemes suffer from limitations in coverage, information timeliness, and scheduling efficiency. To address these challenges, this paper proposes an Integrated Communication-Computing-Intelligence-Controlling (ICCIC) network based on layered uncrewed aerial vehicles (UAVs). The system comprises mobile edge servers (ESs) with trajectory control and scheduling capabilities, and statically deployed working UAVs responsible for wireless channel measurements. The collected data is offloaded to ESs for edge computing and local model construction, which supports subsequent DT channel modeling. A multi-objective optimization problem is formulated to jointly schedule UAV trajectories and data offloading, with Age of Information (AoI) and service delay as performance metrics. The optimization is subject to constraints including maximum task execution time and service fairness. To capture multi-type interactions among UAVs during sensing and coordination, we build a multi-relational heterogeneous graph and encode it with a graph neural network (GNN) to obtain structured system states. On this basis, we adopt a QMIX-based multi-agent reinforcement learning (MARL) framework under centralized training and decentralized execution (CTDE) to learn cooperative scheduling policies. Simulation results demonstrate that the proposed method achieves superior performance in sensing timeliness, scheduling responsiveness, and policy convergence, providing robust support for efficient and reliable DT channel modeling.**

***Index Terms*—Digital twin channel, layered UAVs, trajectory control, data offloading, heterogeneous GNN, MARL, AoI.**

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I. INTRODUCTION

THE evolution of sixth-generation (6G) mobile communication systems is advancing the multi-domain integration of communication, computing, sensing, and control technologies. Future networks will encounter complex channel environments, diverse communication needs, and smarter terminal interactions, supporting dynamic settings for intelligent devices and ubiquitous connectivity [1], [2]. Within this trend, Digital Twin (DT) networks have emerged as a pivotal paradigm for real-time physical–virtual mapping, providing essential capabilities for environment modeling and intelligent network management [3], [4]. To maintain the accuracy and consistency of these mappings, continuous updates of wireless channel conditions are required. Accordingly, DT channel modeling becomes central to bridging the physical and virtual layers, enabling timely and structured representations of dynamic propagation environments. However, in dense urban areas, link conditions are highly non-stationary due to complex spatial structures and frequent obstructions. Traditional methods relying on static measurements or offline data fail to meet the requirements of real-time modeling [5].

Uncrewed aerial vehicles (UAVs), with their high mobility and flexible deployment, offer promising potential for aerial wireless sensing. Compared to static ground nodes, UAVs can actively traverse non-line-of-sight (NLoS) regions and adapt their trajectories to perform localized channel measurements and task-level data collection [6], [7], [8]. Layered UAV systems, in particular, provide spatially distributed coverage and coordinated sensing capabilities, allowing persistent and multi-scale monitoring of dynamic wireless environments. These advantages enable DT channel modeling to shift from static, offline reconstruction to continuous and real-time updates [9].

Despite the advantages of flexible deployment and wide-area coverage, layered UAV systems face multiple challenges when applied to DT channel modeling. The wireless channels in these scenarios are highly unstable, with frequent LoS/NLoS transitions and significant sensitivity to trajectory deviations, making accurate modeling difficult [10]. Additionally, the limited sensing range of individual UAVs often leads to redundant overlaps and sensing voids in cooperative deployments, resulting in uneven data distribution and increased resource waste [11], [12]. On a system level, multi-UAV operations involve simultaneous trajectory planning, data acquisition, and task execution, which are difficult to coordinate efficiently using traditional decoupled or static optimization approaches [13], [14] [15].

These challenges expose two fundamental limitations of existing methods in supporting DT modeling: a lack of efficiency in jointly coordinating sensing and movement, and weak robustness under dynamic, partially observable network conditions. To quantify and optimize sensing timeliness, the Age of Information (AoI) metric [16] has been adopted in trajectory design [17], energy control [18], and other submodules. However, most prior works address AoI or service latency in isolation, and lack a unified framework that jointly optimizes both metrics within a single scheduling process that coordinates trajectory and sensing operations. Moreover, they typically assume fixed or fully observable network topologies, overlooking the structural variability and information asymmetry present in real-world UAV-enabled systems. These gaps highlight the need for a joint trajectory and sensing scheduling strategy that improves both the efficiency and robustness of DT channel modeling in complex urban environments.

To address the limitations of conventional methods in DT channel modeling, we propose a joint scheduling approach that simultaneously optimizes UAV trajectories and data offloading, aiming to improve both efficiency and robustness. Layered UAV systems offer ideal platforms for such integration, with aerial coverage, mobility, and edge processing capabilities. Guided by this, we design an Integrated Communication-Computing-Intelligence-Controlling (ICCIC) network as the foundation for trajectory control and task coordination, and statically deployed working UAVs for wireless channel measurements. We formulate a timeliness-aware scheduling problem, where AoI and service delay are coupled through a product-form objective under deadline and fairness constraints. To address partial observability and interaction complexity, we introduce a graph-based learning framework that combines state abstraction with multi-agent reinforcement learning (MARL).

The main contributions of this paper are as follows:

1) We introduce a layered UAVs-enabled ICCIC network that coordinates the trajectories of edge servers (ESs) and the data offloading from working UAVs, under constraints such as maximum task completion time and service fairness. The dual optimization objectives aim to minimize AoI and service delay, thereby enabling efficient DT channel modeling.

2) To accurately capture the heterogeneous structural characteristics of scheduling and service interactions among layered UAV nodes, a graph neural network (GNN)-based state representation method is introduced. This method employs multi-relation graph encoding to model the topological dependencies of both cooperative and service edges, constructing structured state embeddings. Such representation strengthens the network's robustness against dynamic topology changes, overlapping sensing regions, and coupled scheduling.

3) Building on this foundation, we develop an HG-MARL (Heterogeneous Graph-based Multi-Agent Reinforcement Learning) joint training framework that integrates trajectory control with offloading scheduling. The framework employs a centralized training and decentralized execution (CTDE) paradigm and leverages the QMIX joint value function to enhance coordination among multiple agents. Furthermore, a multi-objective reward function is designed by incorporating service delay and AoI, which enables efficient policy optimization of UAVs' multi-dimensional behaviors in partially observable environments.

4) Extensive simulations show that the proposed method outperforms representative RL baselines by reducing AoI and average service delay, and achieving faster, more stable convergence. It remains robust under variations in LoS/NLoS conditions and noise levels, thereby supporting DT channel modeling in large-scale dynamic urban scenarios.

II. Related Works

A. Layered UAVs Collaboration Architectures in Digital Twin Networks

DT networks are increasingly applied to environmental modeling and adaptive control in wireless systems [19], [20]. In dense urban scenarios with frequent LoS/NLoS transitions, achieving real-time, fine-grained channel modeling remains difficult, especially in ensuring data timeliness, continuity, and balanced spatial coverage.

To address these issues, recent research has introduced layered UAVs collaboration mechanisms to expand sensing coverage, improve spatial resolution, and accelerate data acquisition for DT updates. Li et al. [21] proposed a DT-based online channel modeling framework that forms a perception–modeling–prediction loop for CSI prediction. Gong et al. [22] developed a diffusion-model-based digital twin of channel for sensing-assisted statistical CSI generation. Jiang et al. [23] developed a near-field channel modeling and analysis method for large-scale RIS-enabled air–ground communications, providing theoretical support for UAV-assisted channel sensing optimization. Li et al. [24] designed a DT-assisted UAV vehicular edge computing system with PPO-based joint trajectory and offloading optimization, while

Li et al. [25] proposed a digital-twin approach for UAV-aided edge networks with ultra-reliable low-latency communications.

Beyond pure channel sensing, several works have incorporated computation and resource scheduling into layered UAVs frameworks to better support DT operations. Li et al. [26] proposed an adaptive DT-driven layered UAVs architecture MADRL for joint optimization of task offloading and radar beam design. Consul et al. [27] introduced a DT–MEC framework with federated RL to optimize offloading and resource sharing under privacy constraints. Related incentive-aware client scheduling has also been studied in communication–sensing–computing integrated UAV inspection, showing that incentive and client selection can directly affect resource usage and learning efficiency [28]. Li et al. [29] presented an aerial edge computing system for joint access, trajectory, and resource allocation, using DDQN and iterative methods to improve energy efficiency. Qi et al. [30] integrated millimeter-wave radar with 3D ray tracing for DT channel modeling, enabling energy-efficient communication and precise sensing in green IoT scenarios. Liang et al. [31] and Khalaf et al. [32] further explored UAV-based DT systems by introducing virtual prediction models and AoDT metrics, enabling synchronized airspace sensing and physical-state updates.

While prior studies have explored layered UAVs for channel modeling in DT networks, they often optimize data timeliness and communication delay in isolation, show limited adaptability to dynamic network topologies, and rarely achieve effective joint scheduling of trajectory planning and data offloading. These limitations motivate our work on a layered UAVs-enabled ICCIC network, in which mobile ESs collaborate with working UAVs to conduct wireless channel measurement, data offloading, and edge computing, thereby improving the efficiency and robustness of DT channel modeling in dynamic urban scenarios.

B. UAV Scheduling Optimization in ICCIC Network

In dynamic DT modeling scenarios, UAVs must not only collect data but also handle computation, communication scheduling, and trajectory control, which has driven the development of multi-domain integrated networks. Wang et al. [33] proposed a unified architecture for sensing–computing–intelligence integration in ubiquitous IoT, focusing on enabling technologies such as multimodal perception, control, and orchestration. Chen et al. [34] studied AoI minimization for data collection by cellular-connected UAVs, while Oubbati et al. [35] investigated a UAV–UGV cooperative system for urban monitoring and energy management. From a security perspective, recent studies on distributed AI-based secure communications in integrated space–air–ground–sea networks suggest that learning-driven scheduling can also incorporate security and privacy risks as additional constraints or penalties [36]. Zhang et al. [37] jointly optimized transmit power and trajectory for UAV-enabled data collection under dynamic constraints, while Yao et al. [38] studied joint UAV trajectory and resource allocation for federated learning.

To enhance adaptability in dynamic environments, MARL methods have been widely applied. Cui et al. [39] developed an independent Q-learning framework for channel and power allocation without state sharing. Wu et al. [40] introduced UAV redeployment to mitigate sensing blind spots. Joint modeling of edge computing and task offloading has also gained attention: Sun et al. [41] optimized task distribution, trajectory, and communication resources in a multi-UAV MEC system; Zeng et al. [42] introduced deep reinforcement learning combined with radio mapping to improve path learning efficiency and navigation stability. He et al. [43] proposed a 3D trajectory optimization method with fairness constraints, using MADDPG to improve coordination among UAVs. Tan et al. [44] introduced a communication-assisted MARL framework for task offloading in UAV-aided edge-computing networks.

Recent studies have explored graph-based modeling and distributed policy learning. Oubbati et al. [45] studied on-demand routing with cooperating UAVs in urban networks, which also highlights scalability issues in large deployments. Yang et al. [46] constructed a channel knowledge graph using binary Bayesian filtering, improving LoS connectivity modeling in cellular environments. Zhang et al. [47] employed graph-based MARL to improve collaborative search and tracking performance in multi-UAV systems. Li et al. [48] studied GNN-based scheduling for multi-UAV-enabled communications in D2D networks, while Zhang et al. [49] developed a cooperative trajectory design method for multiple UAV base stations based on heterogeneous graph neural networks.

Despite these advances, few works systematically integrate DT systems and multi-domain integrated networks to meet the real-time, continuous, and balanced scheduling demands of channel modeling—particularly in highly dynamic urban environments where frequent channel updates are crucial for model accuracy and responsiveness. To bridge this gap, we propose an HG-MARL joint optimization framework that models layered UAVs coordination via a multi-relational heterogeneous graph, extracts structured state embeddings for robust topology and channel representation, and applies QMIX under the CTDE paradigm to jointly optimize trajectory control and data offloading policies.

III. Scenario Description and Framework Design

This section presents a layered UAVs-enabled ICCIC network that supports timely link-state sensing and data offloading for DT communication systems. The architecture comprises two aerial tiers—mobile ESs that perform trajectory control and scheduling, and statically deployed working UAVs that conduct wireless channel measurements. Through coordinated sensing and offloading, the system enables distributed data collection, edge computing on ESs, and subsequent model updates at the ground base station.

A. Scenario Description of Layered UAVs in ICCIC Network

This UAV system includes two types of aerial nodes with distinct roles. Working UAVs are statically positioned at fixed sensing points to perform periodic wireless channel measurements. They monitor local signal features—such as path loss, blockage status, and delay—via signal probes or beacons

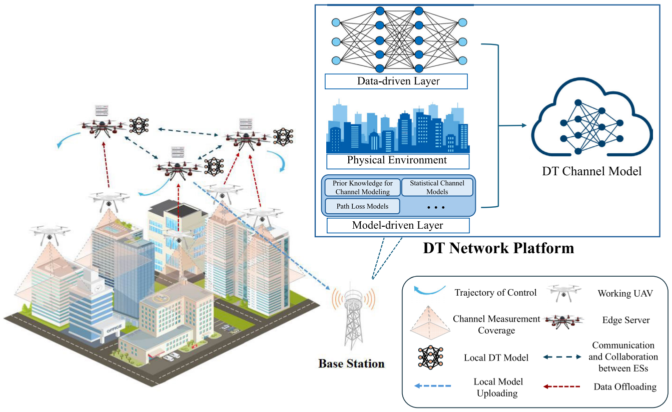


Fig. 1. System scenario diagram.

from user equipment, particularly capturing variations in NLoS regions. Mobile ESs act as data receivers and adjust their trajectories according to RL-based policies. By dynamically moving across the sensing field, they sequentially access multiple working UAVs during each sensing cycle to receive offloaded measurement data. The received data is processed locally to generate intermediate models, which are then used for subsequent DT updates. ESs cruise at fixed altitudes and do not control the deployment of other nodes. This layered design enables scalable and low-latency data offloading. The static working UAVs ensure consistent channel probing at key locations, while ESs support flexible coordination across coverage areas. The system architecture is shown in Fig. 1.

In this system, the set of ESs is denoted as $\mathcal{V}_{ES} = \{1, \ldots, N_{ES}\}$, and the set of working UAVs is defined as $\mathcal{V}_{UAV} = \{1, \ldots, N_{UAV}\}$. For clarity, key notations are: slot duration Δt (s), decision $q_{k,i}(t) \in \{0, 1\}$, total bandwidth B (Hz), transmit power P_{tx} (W), noise σ^2, v_{max} (m/s), $T_i^{\max}$ (slots), and η. See Table I for values.

For modeling purposes, the total system runtime is divided into T discrete time slots, each with duration Δt. Within each time slot, working UAVs remain stationary at their assigned sensing sites, whereas only ESs dynamically adjust their 3D positions according to the scheduling policy to optimize coverage; these ES motions do not modify the locations of working UAVs. It is assumed that all ESs and working UAVs operate within a flight area defined by an $M \times N$ grid. Each ES flies at a specified altitude H_{ES}, while each working UAV hovers at a fixed but lower height H_{UAV}. These heights satisfy the condition $H_{ES} > H_{BMax} > H_{UAV}$ to avoid collisions and ensure effective communication coverage, where H_{BMax} denotes the maximum building height in the environment. Moreover, collisions among mobile ESs are avoided by enforcing a minimum separation distance during operation, which is satisfied in our simulations due to the small ES scale and altitude separation. Let $\mathbf{p}_k^{UAV}(t) = (x_k^{UAV}(t), y_k^{UAV}(t), z_k^{UAV}(t))$ denote the position of the k-th working UAV at time slot t, and $\mathbf{p}_i^{ES}(t) = (x_i^{ES}(t), y_i^{ES}(t), z_i^{ES}(t))$ denote the position of ES_i at time slot t. Together, they form a structured system that supports reliable and fine-grained DT channel modeling in dynamic environments.

TABLE I
SIMULATION PARAMETERS

Parameter (Notation)	Value
3D City Map Dimensions ($M \times N$)	800×600
Building Maximum/Minimum Height (H_{BMax}/H_{BMin})	65m, 15m
Building Average Height ($\bar{H}_B$)	30m
Working UAV Hovering Height (H_{UAV})	30-40m
ES Flying Height (H_{ES})	70-80m
Total Bandwidth (B)	10MHz
ES Transmit Power (P_{tx})	40W
Gaussian Noise (σ^2)	-60dBm
Parameter Synchronization Interval (N_{comm})	5
Number of GNN Layers (L)	2
GNN Hidden Layer Dimension	128
RNN Hidden Layer Dimension	64
Penalty Coefficients ($\alpha_1, \alpha_2, \alpha_3$)	0.05, 0.2, 0.5
Parameter Transfer per Working UAV	5000bits
ES Maximum Flight Speed (v_{max})	10m/s
Per Time Step Duration (Δt)	1s
Maximum Steps per Episode ($T_i^{\max}$)	100
Fairness Threshold (η)	0.9
Training Episodes (N_{ep})	10000
Initial Exploration Rate	0.5
Final Exploration Rate	0.05
Learning Rate (η_q)	0.0005
Learning Rate (η_ϕ, GNN encoder)	0.0001
Batch Size	32
Experience Replay Buffer Size	5000
Discount Factor (γ)	0.9

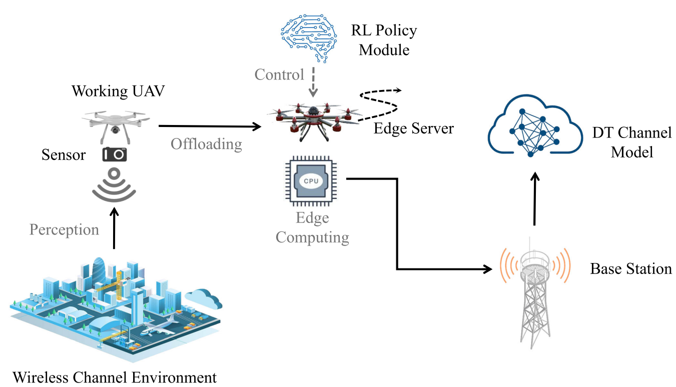


Fig. 2. Functional architecture of layered UAVs-enabled ICCIC network.

B. Joint Scheduling of Trajectory and Offloading for Digital Twin Online Channel Modeling

To enable efficient and robust modeling of the wireless propagation environment in DT networks, this paper proposes a joint process that integrates dynamic trajectory adjustments with real-time data offloading, ensuring timely processing and high-quality inputs for the DT model.

In each task cycle, working UAV k hovers at a fixed altitude H_{UAV} and performs periodic wireless channel measurements within its assigned area using onboard sensing equipment. Meanwhile, a mobile edge server ES_i flies at altitude H_{ES} along an adaptive 3D trajectory with speed

$v_i(t) = \left\|\mathbf{p}_i^{ES}(t+1) - \mathbf{p}_i^{ES}(t)\right\|_2 / \Delta t$. Its movement and scheduling decisions are determined by a reinforcement learning policy, enabling it to sequentially visit working UAVs, receive offloaded measurement data, and conduct edge computing for intermediate channel modeling. Fig. 2 illustrates the ICCIC workflow, which integrates UAV-based sensing, wireless data offloading, edge computing, and scheduling control.

In the overall architecture, computational tasks are primarily executed by the ESs. After receiving measurement data from multiple UAVs, each ES generates an intermediate channel model to preserve representative channel characteristics while reducing transmission overhead. At the end of each task cycle, the system selects an ES node with the largest remaining task-time budget (as a proxy of residual energy under $T_i^{\max}$) to act as the relay for model uploading; if multiple ESs have similar remaining budgets, the one with lower uploading latency is chosen. This selected node independently transmits its intermediate model to a remote ground base station, serving as input for subsequent DT channel modeling.

Upon receiving the uploaded data, the base station invokes a pre-existing channel modeling module to either build a new model or update an existing one. This module adopts a three-layer DT architecture that integrates data-driven learning, spatial environment encoding, and model-based priors into a unified pipeline to generate the DT channel model. It forms a closed loop of measurement, offloading, modeling, and optimization, enhancing responsiveness and control under dynamic channel conditions.

IV. System Model and Problem Formulation

To ensure the efficient execution of the proposed network, the system must coordinate several interdependent processes, including wireless communication, channel measurement, data offloading with edge computing, and scheduling control. These tasks exhibit complex spatiotemporal dependencies, necessitating a unified modeling approach to capture the dynamic interactions and structural constraints among various entities. This section develops a mathematical formulation for the ICCIC network and formally defines the joint optimization.

A. Communication and Offloading Model

In the proposed network, data offloading from working UAVs to ESs requires the establishment of a viable wireless link [49]. The probability that ES_i successfully connects to UAV_k at time step t is denoted by $P_{k,i}^{conn}(t)$ and is given by:

$$P_{k,i}^{conn}(t) = P_{k,i}^{LoS}(t) \cdot e^{-\beta d_{k,i}(t)}. \tag{1}$$

Let $d_{k,i}(t)$ denote the Euclidean distance between UAV_k and ES_i at time step t, and let β represent the path-loss attenuation factor. The probability of a LoS link between ES_i and UAV_k is denoted as $P_{k,i}^{LoS}(t)$, and it is calculated as follows:

$$P_{k,i}^{LoS}(t) = \frac{1}{1 + a \cdot \exp(-b(\theta_{k,i} - a))}, \tag{2}$$

where $\theta_{k,i}$ is the elevation angle, and a, b are path-loss related parameters associated with the urban environment. Once a link is available, the ES allocates a sub-channel to the UAV via a Frequency Division Multiple Access (FDMA) scheme, where the total bandwidth B is divided into N_c sub-channels and each UAV occupies one dedicated channel. A binary variable $q_{k,i}(t) \in \{0,1\}$ is introduced to indicate whether UAV_k is scheduled by ES_i at time step t, where $q_{k,i}(t) = 1$ means that ES_i allocates a communication sub-channel to UAV_k for offloading its measured channel data, and $q_{k,i}(t) = 0$ means that UAV_k is not scheduled by ES_i at that time step.

The transmit power of each ES on each sub-channel is fixed at P_{tx}. To further reduce interference and improve communication efficiency, the ESs are equipped with directional antennas that emit signals within a fixed beamwidth angle θ. Consequently, the coverage radius R_c is determined by: $R_c = |H_{ES} - H_{UAV}| \cdot \tan\theta$. Under different propagation conditions, the channel gain between the UAV and ES varies. In the case of LoS transmission, the channel gain is given by:

$$g_{k,i}^{LoS}(t) = \frac{h_0}{(d_{k,i}(t))^{\alpha}}. \tag{3}$$

When propagating with NLoS [50], the channel gain needs to be corrected by adding an additional path loss factor η_{NLoS} due to the signal multipath effect and building obstruction:

$$g_{k,i}^{NLoS}(t) = \eta_{NLoS} \cdot g_{k,i}^{LoS}(t), \tag{4}$$

where h_0 is the reference path loss constant and α is the path loss exponent, the channel gain $g_{k,i}(t)$ between ES_i and UAV_k at time step t can be expressed as:

$$g_{k,i}(t) = P_{k,i}^{LoS}(t) \cdot g_{k,i}^{LoS}(t) + (1 - P_{k,i}^{LoS}(t)) \cdot g_{k,i}^{NLoS}(t). \tag{5}$$

This LoS/NLoS large-scale model captures the average attenuation and is widely adopted in UAV placement/trajectory optimization studies; small-scale fading and Doppler effects are not explicitly modeled here, which fits slot-based scheduling using per-slot averaged link quality [51], [52], [53]. We define the coverage relation matrix $C_{k,i}(t) \in \{0,1\}$, where $C_{k,i}(t) = 1$ indicates that the signal of ES_i can effectively cover UAV_k at time step t. For any time step t, the received signal power of UAV_k can be expressed as if $C_{k,i}(t) = 1$:

$$P_{k,i}(t) = C_{k,i}(t) \cdot g_{k,i}(t) \cdot P_{tx}. \tag{6}$$

Further calculate the instantaneous throughput of UAV_k at time step t:

$$R_k(t) = W \cdot \log_2\left(1 + \frac{P_{k,i}(t)}{\sigma^2 + \sum_{j \neq i} \sum_{k' \in \mathcal{K}_j(t)} P_{k',j}(t)}\right), \tag{7}$$

where $W = B/N_c$ denotes the bandwidth of each dedicated sub-channel under FDMA, $\mathcal{K}_j(t)$ denotes the set of UAVs served by ES_j at time step t. In the DT channel construction task, we define $Data_k(t)$ as the total amount of link-aware data uploaded by UAV_k in a single time step. The transmission delay $D_k(t)$ is used to measure the time required for these data transmissions and is calculated as:

$$D_k(t) = \frac{Data_k(t)}{R_k(t)}. \tag{8}$$

To quantify the timeliness of UAV-reported information, this paper introduces the AoI as a performance metric [54]. For

ES_i, after receiving the latest link observation from UAV_k at time slot t, the AoI is defined as:

$$AoI_{k,i}(t) = t - \zeta_{k,i}(t), \tag{9}$$

where $AoI_{k,i}(t)$ represents the elapsed time since the most recent data from UAV_k was successfully received by ES_i, and $\zeta_{k,i}(t)$ denotes the time slot of that last successful reception.

Note that the denominators in (10) and (15) count the scheduled UAV–ES associations through $\sum_{k=1}^{N_{UAV}} \sum_{i=1}^{N_{ES}} q_{k,i}(t)$. In our simulations, at least one feasible UAV–ES pair is scheduled in each time slot, so $\sum_{k,i} q_{k,i}(t) > 0$ and division-by-zero does not occur. To evaluate data update quality per task cycle, we define a system-level AoI. This metric measures the average timeliness degradation of UAV measurement data received by ESs during the cycle. It is calculated as:

$$\overline{AoI}_{sys} = \frac{1}{T} \sum_{t=1}^{T} \frac{\sum_{k=1}^{N_{UAV}} \sum_{i=1}^{N_{ES}} q_{k,i}(t) \cdot AoI_{k,i}(t)}{\sum_{k=1}^{N_{UAV}} \sum_{i=1}^{N_{ES}} q_{k,i}(t)}. \tag{10}$$

Similarly, to prevent UAVs from remaining in service blind zones, we introduce a system-level service fairness metric. Since ESs can only serve a limited number of dispersed UAVs at any time, optimal trajectories may require temporary unfairness. Therefore, we adopt a long-term fairness perspective to ensure balanced service across the task horizon. The average throughput of UAV_k before time step t is defined as $\bar{R}_k(t) = \frac{1}{t} \sum_{t'=1}^{t} R_k(t')$. The set of average throughputs for all UAVs is denoted as $\overline{R}_t = \left[\bar{R}_1(t), \ldots, \bar{R}_{N_{UAV}}(t)\right]^{\mathrm{T}}$.The fairness among different UAVs up to time step t is measured by the Jain's Fairness Index, which is defined as:

$$\boldsymbol{J}\left(\overline{R}_t\right) = \frac{\left(\sum_{k=1}^{N_{UAV}} \bar{R}_k(t)\right)^2}{N_{UAV} \sum_{k=1}^{N_{UAV}} \bar{R}_k(t)^2}. \tag{11}$$

The value of $\boldsymbol{J}\left(\overline{R}_t\right)$ lies in the range $[1/N_{UAV}, 1]$, where a value closer to 1 indicates more balanced communication opportunities among UAVs, and hence, higher service fairness in the system. In our formulation, fairness is enforced by the constraint C6 in P1 and is also encouraged during learning by the soft penalty term in the reward (34).

B. Computation and Control Model

Since working UAVs focus solely on channel measurement and lack computational and modeling capabilities, the raw measurement data they sense must be offloaded to the ES that covers their service area. Upon receiving the data, the ES performs edge computation to construct a local channel model and carry out subsequent processing tasks. The total aggregated data volume is:

$$Data_i^{agg}(t) = \sum_{k=1}^{N_{UAV}} q_{k,i}(t) \cdot Data_k(t). \tag{12}$$

Let the computational complexity per bit of data be denoted by $c_i(t)$ and the processing capability of ES node i be $f_i(t)$. Then, the local computation delay is given by:

$$D_i^{comp}(t) = \frac{c_i(t) \cdot Data_i^{agg}(t)}{f_i(t)}. \tag{13}$$

The Average Service Delay metric is adopted to comprehensively evaluate the system-level time consumption in the task processing chain. For ES_i receiving and processing the offloaded measurement data from UAV_k at time slot t, the instantaneous service delay is defined as:

$$D_{k,i}^{serv}(t) = D_k(t) + D_i^{comp}(t). \tag{14}$$

The average service delay over the entire task period T is given by:

$$\overline{D}_{serv} = \frac{1}{T} \sum_{t=1}^{T} \frac{\sum_{k=1}^{N_{UAV}} \sum_{i=1}^{N_{ES}} q_{k,i}(t) \cdot D_{k,i}^{serv}(t)}{\sum_{k=1}^{N_{UAV}} \sum_{i=1}^{N_{ES}} q_{k,i}(t)}. \tag{15}$$

In terms of the control model, each ES node possesses flexible trajectory adjustment capability. Its control actions include the position control vector $\mathbf{u}_i(t)$ and the data offloading scheduling variable $q_{k,i}(t)$. The action set $\mathcal{A}_i$ consists of five discrete actions: move north, move south, move east, move west, and hover, each corresponding to a fixed step size Δd update in the two-dimensional space, formulated as:

$$\mathbf{u}_i(t) \in \mathcal{A}_i = \{(0, \Delta d), (0, -\Delta d), (\Delta d, 0), (-\Delta d, 0), (0, 0)\}. \tag{16}$$

Accordingly, the position update of ES_i is expressed as:

$$\mathbf{p}_i^{ES}(t+1) = \mathbf{p}_i^{ES}(t) + \mathbf{u}_i(t). \tag{17}$$

C. Problem Formulation

We propose a system-level policy design method that jointly optimizes trajectory control and offloading within the layered UAVs cooperative framework. Given the time-critical nature of channel data updates and the uneven distribution of service resources across nodes, the optimization objective is defined as minimizing the cumulative product of instantaneous AoI and service delay over the entire task period. The product form, rather than a linear sum, is chosen to better capture the coupling between information staleness and transmission latency, highlighting bottlenecks in either dimension. This formulation guides mobile ESs to generate scheduling policies that are both timely and stable in dynamic environments.

For the constraint design, the system must consider multiple feasibility conditions for joint scheduling. A lower bound threshold of long-term fairness, denoted as η, is introduced as a hard constraint in the overall optimization model to ensure that the communication performance is improved without compromising the fair allocation of system resources. Moreover, since the ES nodes have limited energy capacity, their continuous flight and execution of tasks cannot exceed a certain duration. To prevent energy depletion, each ES node is restricted by a maximum task period $T_i^{\max}$. This time-budget acts as a simple proxy for energy management: more aggressive movement/serving may reduce AoI and delay, but it is limited by $T_i^{\max}$. In this work, the working UAVs are assumed to keep hovering at fixed sensing sites with sufficient onboard energy, and a detailed UAV battery model is left for future extensions. Based on the above considerations, the formulated optimization problem is given as follows:

$$
\begin{aligned}
\mathrm{P1:}\ &\min_{\{\mathbf{a}_i(t)\}_{i\in\mathcal{V}_{ES}}^{T}} \frac{1}{T}\sum_{t=1}^{T}\sum_{k=1}^{N_{UAV}}\sum_{i=1}^{N_{ES}} q_{k,i}(t)\cdot AoI_{k,i}(t)\cdot D_{k,i}^{serv}(t)\\
&\text{s.t. C1:}||\mathbf{p}_i^{ES}(t+1)-\mathbf{p}_i^{ES}(t)||_2/\Delta t \le v_{max}, \quad \forall i\in\mathcal{V}_{ES},\\
&\quad \text{C2:}x_i^{ES}(t)\in[0,M], \quad y_i^{ES}(t)\in[0,N], \quad \forall i\in\mathcal{V}_{ES},\\
&\quad \text{C3:}\sum_{i=1}^{N_{ES}} q_{k,i}(t)\le 1, \quad \forall k\in\mathcal{V}_{UAV},\\
&\quad \text{C4:}\sum_{k=1}^{N_{UAV}} q_{k,i}(t)\le N_c, \quad \forall i\in\mathcal{V}_{ES},\\
&\quad \text{C5:}T\le T_i^{\max}, \quad \forall i\in\mathcal{V}_{ES},\\
&\quad \text{C6:}\boldsymbol{J}\left(\overline{R}_t\right)\ge\eta,\\
&\quad \text{C7:}\left\|\mathbf{p}_i^{ES}(t)-\mathbf{p}_j^{ES}(t)\right\|_2\ge d_{\min},\ \forall i\ne j, i,j\in\mathcal{V}_{ES},
\end{aligned} \tag{18}
$$

where $\mathbf{a}_i(t)=(\mathbf{u}_i(t),q_{k,i}(t))$, and $\mathcal{A}_i$ denotes the corresponding action space. C1 limits the maximum flying speed of each ES; C2 ensures that all ESs remain within the designated operation area; C3 stipulates that each UAV can be served by at most one ES on a single sub-channel; C4 limits the number of UAVs served by each ES to not exceed the available number of sub-channels; C5 sets an upper bound on the total task completion time of the entire scheduling process; C6 guarantees the fairness of resource allocation across the system; C7 enforces a minimum safety separation among ESs.

In implementation, C1–C5 and C7 are enforced through the action space and feasibility checks by masking infeasible moves and associations, while C6 is encouraged by a fairness penalty in the reward and monitored during training and evaluation. This framework can be extended to cross-layer settings by augmenting the state with application- and transport-layer information and by incorporating end-to-end QoS targets as additional constraints or reward terms.

It is important to note that the proposed joint scheduling problem is inherently non-convex, primarily because the objective function is defined as the product of AoI and service delay. The inclusion of fairness, trajectory, and energy constraints further enlarges the solution space, making traditional convex optimization methods inefficient. To tackle this challenge, we adopt a value-based MARL approach, leveraging a GNN-enhanced QMIX framework under centralized training and decentralized execution to achieve efficient policy optimization.

V. Heterogeneous Graph-Based State Representation and Structural Modeling Design

In the collaborative tasks of the layered UAVs-enabled ICCIC network, there exist complex and dynamically changing interactions among nodes. Traditional state modeling approaches are insufficient to accurately capture such unstructured spatial relationships, especially when local information is incomplete and heterogeneous collaboration exists between nodes. To better support the optimization of collaborative control strategies in layered UAV systems, this section introduces a state modeling method based on GNN. By leveraging the structure of a heterogeneous graph, this method effectively represents the states and interactions among nodes, thereby improving the quality of information available for system-level decision-making.

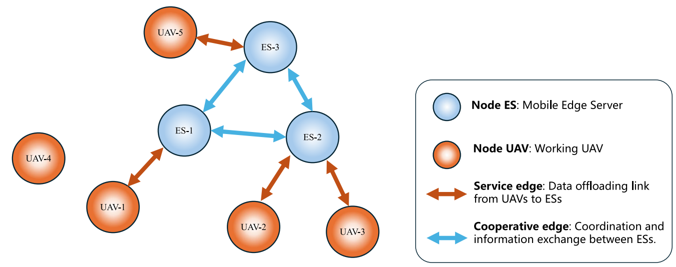


Fig. 3. Heterogeneous graph structure for ES–UAV interaction.

A. Multi-Relational Heterogeneous Graph Structure Definition

To achieve structured modeling for the tasks in the DT channel scenario, a multi-relational heterogeneous graph is introduced during the system state encoding stage. This structure supports dynamic interactions among different types of nodes, making it suitable for modeling service connections and cooperative exchanges in multi-UAV collaboration. Unlike conventional homogeneous graphs that handle only a single type of node and edge, the heterogeneous graph simultaneously represents the scheduling relationships between ESs and UAVs and the communication coordination among ESs, providing structured semantic input for downstream GNN-based state extraction. Fig. 3 illustrates the heterogeneous graph structure used to model the interaction between them.

At each time slot t, the system constructs a heterogeneous graph $\mathcal{G}_t=\{\mathcal{V}_t,\mathcal{E}_t\}$, where the node set $\mathcal{V}_t=\mathcal{V}_t^{ES}\cup\mathcal{V}_t^{UAV}$ includes two types of nodes: ES nodes and service-requesting UAV nodes. The edge set $\mathcal{E}_t$ contains two types of relational edges to capture task-related spatial interactions.

The first type is the service edge, which connects an ES node with the UAVs located within its sensing range and represents a potential data offloading link from UAVs to ESs. The establishment of such an edge depends on LoS availability, geographical distance, and channel quality. If LoS is present and the link satisfies the minimum SINR requirement, a valid service edge $\mathbf{e}_{ik}^{\text{serv}}\in\mathcal{E}_t$ is created, with associated features including SINR, distance, and LoS indicators.

The second type is the cooperative edge, which connects two ES nodes and captures their spatial adjacency and potential coverage overlap. A cooperative edge $\mathbf{e}_{ij}^{\text{coop}}\in\mathcal{E}_t$ is established when the two ESs meet the cooperation condition, such as overlapping service regions or possible scheduling conflicts. These edges provide the structural basis for localized cooperation, enabling neighboring ESs to exchange information through the graph and supporting stable parameter synchronization during multi-agent training.

B. Heterogeneous GNN Input Design

To effectively model the heterogeneous information in the graph, the system designs specific feature encoding schemes for both node types and edge types. This allows the GNN to distinguish between different roles of nodes, identify the semantic meaning of different types of relationships, and subsequently perform targeted information aggregation.

For node features, we distinguish between the two types of nodes in the graph and construct attribute vectors separately. For an ES node $v_i \in \mathcal{V}_t^{ES}$, its input feature vector $\mathbf{x}_i^t \in \mathbb{R}^{d_{ES}}$ is constructed as follows:

$$\mathbf{x}_i^t = [\mathbf{p}_i^t, \rho_i^t, \theta_i^t], \tag{19}$$

where $\mathbf{p}_i^t \in \mathcal{D}$ denotes the spatial coordinate of ES, ρ_i^t represents the available bandwidth resource, and θ_i^t indicates the number of current connections, which reflects the load status of the ES node. For a UAV node $v_k \in \mathcal{V}_t^{UAV}$, its input feature vector $\mathbf{x}_k^t \in \mathbb{R}^{d_{UAV}}$ is defined as:

$$\mathbf{x}_k^t = \left[\mathbf{p}_k^t, \phi_k^t, \omega_k^t\right], \tag{20}$$

where $\phi_k^t \in \{0,1\}$ denotes whether the UAV has data to upload, and ω_k^t is the cumulative time it has remained in the service state.

For edge features, we define attribute vectors for both service edges and cooperative edges. For a service edge $\mathbf{e}_{ik}^{serv} \in \mathcal{E}_t$, its feature encoding is given by:

$$\mathbf{e}_{ik}^{serv} = \left[d_{ik}^t, SINR_{ik}^t, LoS_{ik}^t\right], \tag{21}$$

where d_{ik}^t is the spatial distance, $SINR_{ik}^t$ is the signal-to-interference-plus-noise ratio of the link, and $LoS_{ik}^t \in \{0,1\}$ indicates whether there is a LoS condition between the ES and UAV. For a cooperative edge $e_{ij}^{coop} \in \mathcal{E}_t$, its edge feature is defined as:

$$\mathbf{e}_{ij}^{coop} = [d_{ij}^t, \xi_{ij}^t], \tag{22}$$

where d_{ij}^t is the spatial distance between two ES nodes, and $\xi_{ij}^t \in \{0,1\}$ is a cooperation flag indicating whether the cooperation condition is met, for example, due to overlapping coverage areas or potential scheduling conflicts.

At each time step, the graph $\mathcal{G}_t$ is updated based on the current geometry and link feasibility. Specifically, we add a service edge only if the ES–UAV link is available under the LoS/NLoS model and its SINR meets the decoding requirement; cooperative edges are added when the cooperation condition is satisfied. Notably, heterogeneous capabilities (e.g., range/sensing limits) can be modeled by adding capability/QoS features to nodes and edges and masking infeasible actions; for much stronger heterogeneity, more flexible MARL designs can be considered in future work. To enable the GNN to distinguish the roles of different edge types during information propagation, we adopt an edge-type-specific encoding mechanism as introduced in [52]. Specifically, during message passing, we apply edge-type-specific transformations. The node embedding update at layer $l+1$ is written as:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{r\in\mathcal{J}}\sum_{j\in\mathcal{N}_i^r} \alpha_{ij}^{(r)} \boldsymbol{W}_r \mathbf{h}_j^{(l)}\right), \tag{23}$$

where $\mathcal{J} = \{serv, coop\}$ denotes the set of edge types, $\alpha_{ij}^{(r)}$ is the attention weight, $\boldsymbol{W}_r$ is the message transformation matrix for edge type, and $\mathbf{h}_j^{(l)}$ is the l layer embedding of node j. The function $\sigma(\cdot)$ represents a non-linear activation function, such as *LeakyReLU*, which enhances the network's ability to model complex heterogeneous relationships and edge-feature coupling effects.

In implementation, we use a L-layer message-passing encoder with hidden dimension 128 (Table I) and LeakyReLU activation. The dynamic graph $\mathcal{G}_t$ is updated at each time step based on the current link feasibility and the cooperation flag defined above. Other training hyper-parameters follow the standard QMIX setup and are fixed across all methods.

C. Node State Aggregation and Update Mechanism Based on Heterogeneous Graphs

To enhance the representation capability and structural perception, we propose a message aggregation mechanism that distinguishes edge types. Specifically, service and cooperative edges are processed through separate feature mapping functions to capture attributes such as link quality, scheduling potential, and spatial proximity. During aggregation, an attention-based weighted fusion strategy is applied, enabling each ES agent to adaptively adjust the contribution of neighbor features based on edge types. This design improves the accuracy and discriminability of node embeddings.

1) Multi-Relational Edge Message Encoding and Feature Construction: For different types of edges in the set $\mathcal{E}_t$, we construct separate message passing channels, enabling the GNN to independently process information from various relation types and incorporate the results into ES node state updates. Specifically, for service edges $\mathbf{e}_{ij}^{serv}$, the message passing mechanism is defined as follows:

$$\mathbf{m}_{ik}^{serv} = \boldsymbol{W}_{serv} \cdot \sigma\left(\mathbf{x}_k^t \oplus \mathbf{e}_{ik}^{serv}\right). \tag{24}$$

Here, $\mathbf{x}_k^t$ denotes the feature of UAV node k, $\mathbf{e}_{ik}^{serv}$ represents the feature of the service edge, and $\boldsymbol{W}_{serv}$ is the mapping matrix for service edge messages. $\oplus$ denotes the feature concatenation operation. For cooperative edges $\mathbf{e}_{ij}^{coop}$, different message parameters are used:

$$\mathbf{m}_{ij}^{coop} = \boldsymbol{W}_{coop} \cdot \sigma\left(\mathbf{x}_j^t \oplus \mathbf{e}_{ij}^{coop}\right). \tag{25}$$

These two mechanisms respectively model the service and cooperation relationships, ensuring that the ES node can distinguish heterogeneous message types when updating its state representation.

2) Attention-Based Weighted Message Aggregation: During the message aggregation phase, to address the uneven influence of different neighboring UAVs and ES nodes on policy learning, we introduce an attention mechanism based on edge types to adaptively weight the incoming messages. Specifically, during message passing, the attention weights for service edges and cooperative edges are computed as:

$$\alpha_{ik}^{serv} = \frac{\exp(\mathit{LeakyReLU}(\mathbf{a}_{serv}^T \cdot [\mathbf{x}_i^t \oplus \mathbf{x}_k^t \oplus \mathbf{e}_{ik}^{serv}]))}{\sum_{k'\in\mathcal{N}_i^{serv}} \exp(\mathit{LeakyReLU}(\mathbf{a}_{serv}^T \cdot [\mathbf{x}_i^t \oplus \mathbf{x}_{k'}^t \oplus \mathbf{e}_{ik'}^{serv}]))}, \tag{26}$$

$$\alpha_{ij}^{coop} = \frac{\exp(\mathit{LeakyReLU}(\mathbf{a}_{coop}^T \cdot [\mathbf{x}_i^t \oplus \mathbf{x}_j^t \oplus \mathbf{e}_{ij}^{coop}]))}{\sum_{j' \in \mathcal{N}_i^{coop}} \exp(\mathit{LeakyReLU}(\mathbf{a}_{coop}^T \cdot [\mathbf{x}_i^t \oplus \mathbf{x}_{j'}^t \oplus \mathbf{e}_{ij'}^{coop}]))}, \tag{27}$$

where $\mathbf{a}_{serv}^T$ and $\mathbf{a}_{coop}^T$ are the learned edge-type-aware attention vectors, and α_{ik}^{serv} and α_{ij}^{coop} represent the attention weights for service and cooperative edges, respectively.

3) Node State Update and Message Integration: After obtaining the attention weights, each ES node updates its representation by aggregating messages from neighbors through a weighted sum. For ES node v_i, the layer-wise update is

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{k \in \mathcal{N}_i^{\text{serv}}} \alpha_{ik}^{\text{serv}} \cdot \mathbf{m}_{ik}^{\text{serv}} + \sum_{j \in \mathcal{N}_i^{\text{coop}}} \alpha_{ij}^{\text{coop}} \cdot \mathbf{m}_{ij}^{\text{coop}}\right). \tag{28}$$

This operation aggregates messages from the two neighbor sets and then applies the activation function $\sigma(\cdot)$ to generate the updated embedding $\mathbf{h}_i^{(l+1)}$. The resulting $\mathbf{h}_i^{(l+1)}$ serves as the local representation of v_i at the $(l+1)$-th layer. After recursively applying this update across L layers on the graph at time t, the final embedding $\mathbf{h}_i^t$ is derived and used as the local state input for subsequent MARL-based policy modeling.

VI. Efficient and Robust Strategy Optimization for Layered UAVs

To address collaborative data offloading and scheduling in multi-ES urban scenarios, this paper proposes a joint optimization strategy named HG-MARL, which integrates GNN with a value decomposition mechanism.

A. Policy Modeling for MARL: State Inputs, Joint Actions, and Instant Reward Mechanism

This section introduces the modeling of states, joint actions, and reward mechanisms, which serve as the input foundation for the subsequent GNN-based encoding and QMIX-based joint training. In the proposed MARL framework, each ES is treated as an autonomous agent, indexed by $a_i \in \{1, \ldots, N_{ES}\}$. At each time step, agent a_i makes decisions based on its local observation and the exchanged information with neighboring agents.

1) State: At time step t, the local state of a_i is defined as the output embedding of an L-layer heterogeneous GNN operating on $\mathcal{G}_t$:

$$\mathbf{h}_i^t = \phi_\theta(\mathbf{o}_i^t, \mathcal{G}_t) \in \mathbb{R}^{d_h}, \tag{29}$$

where $\mathbf{o}_i^t$ is the local observation of a_i, $\mathcal{G}_t$ is the graph at time t, and $\phi_\theta(\cdot)$ denotes the L-layer graph encoder. The global state is constructed by concatenating the embeddings of all ES nodes:

$$\mathbf{s}^t = [\mathbf{h}_1^t,\ \mathbf{h}_2^t,\ \ldots,\ \mathbf{h}_{N_{ES}}^t]. \tag{30}$$

During training, $\mathbf{s}^t$ serves as the input to the QMIX network for global value estimation and joint policy optimization.

2) Action: In this system, each agent is required to perform two core control tasks at every time step. One is to adjust its movement trajectory to enhance sensing coverage; the other is to decide whether to support the data uploading from working UAVs currently within its sensing range. For trajectory control, the movement action of each agent is defined as:

$$\mathcal{M} = \{\mathit{Up}, \mathit{Down}, \mathit{Left}, \mathit{Right}, \mathit{Stay}.\}. \tag{31}$$

This design allows flexible trajectory planning and risk-aware control of communication interruptions, helping avoid frequent and severe position changes that may destabilize the network. In the scheduling part, the ES needs to decide whether to support the model uploading request from the UAV nodes within its sensing range. We define the scheduling action of ES_i as a set-based decision:

$$\delta_i^t = \left\{k \in \mathcal{N}_i^{UAV} \,\middle|\, \mathit{selected}(k) = 1\right\}. \tag{32}$$

Here, $\mathcal{N}_i^{UAV}$ represents the candidate UAV nodes currently within the sensing range of ES_i, and $\mathit{selected}(k)$ is a binary decision variable. The decision must satisfy the following constraints: (1) only select UAVs under LoS condition; (2) the number of selected UAVs must not exceed the maximum access capacity of the ES, denoted as N_c. Combining both trajectory and scheduling parts, we define the joint action of ES_i as:

$$\mathbf{a}_i^t = (\mathbf{m}_i^t, \delta_i^t), \quad \mathbf{m}_i^t \in \mathcal{M},\ \delta_i^t \subseteq \mathcal{N}_i^{UAV}. \tag{33}$$

The joint action space exhibits both combinatorial and sparse characteristics. To handle this, we adopt a multi-head architecture in the policy network, where trajectory and scheduling decisions are modeled separately and jointly optimized through their corresponding Q-values. During training, policy fusion is achieved via joint sampling or attention mechanisms, ensuring coordinated spatiotemporal responses between movement and service actions.

3) Reward: The proposed task exhibits significant partial observability and decentralized characteristics, as each agent makes decisions solely based on its local and neighborhood information without access to the global system state. When a UAV is out of range or a transmission link is temporarily unavailable, the corresponding ES–UAV relation is not observed in the local graph and will be updated once the link becomes available again. In addition, since AoI is defined based on the last successful reception time slot, it naturally accumulates during such outages and provides a simple memory of stale information. Accordingly, we design a system-level reward function to guide agents in learning coordinated strategies under local observations [55].

To optimize timeliness and ensure consistency with key performance requirements, we adopt a time-slot-based instantaneous reward function during training. By combining the AoI and service delay metrics with the scheduling variable as a weight, and further incorporating soft penalties on fairness and task completion time, the total reward at each time slot is defined as:

$$R_t = -\sum_{k=1}^{N_{UAV}} \sum_{i=1}^{N_{ES}} q_{k,i}(t) \cdot AoI_{k,i}(t) \cdot D_{k,i}^{\text{serv}}(t) - \alpha_1 \sum_{i=1}^{N_{ES}} \mathbb{I}(p_i^t \notin \mathcal{D}) - \alpha_2 [\eta - \boldsymbol{J}(\overline{R}_t)]_+$$

$$- \alpha_3 \, \mathbb{I}(t = T') \cdot \sum_{i=1}^{N_{ES}} \left[T' - T_i^{\max} \right]_+ . \quad (34)$$

Here, $\mathcal{D}$ denotes the boundary of the permissible two-dimensional flight area, and $\mathbb{I}(\cdot)$ is the indicator function. The first term optimizes timeliness by combining AoI and delay. The second term imposes a penalty when an agent moves beyond the map boundary. The third term introduces a fairness penalty through the Jain's index $\boldsymbol{J}\left(\overline{R}_t\right)$ with threshold η, and the fourth term adds a penalty if the task duration exceeds the maximum limit $T_i^{\max}$. The penalty coefficients are determined by preliminary tuning: we first increase each coefficient until the corresponding violations become rare, and then keep the smallest values that maintain feasibility to avoid over-penalizing the main objective. The constants α_1, α_2, and α_3 are selected by a small grid search under the default simulation configuration to keep the penalty terms comparable in scale to the main timeliness term, and the reported values give the best overall performance among the tested candidates. In particular, α_1 is chosen as the smallest value that discourages boundary violations, α_2 is tuned to meet the fairness threshold η without dominating the reward, and α_3 is used to penalize deadline violations only when $t = T'$. The resulting values are given in Table I. This reward is shared by all agents as a global signal and is used by the QMIX value function module for coordinated policy optimization. Overall, the first term directly follows our goal of reducing the AoI–delay product, while the other terms act as soft constraints; in practice, the penalties are set to discourage boundary, fairness, and deadline violations without dominating the timeliness objective.

B. Joint Multi-Agent Training Framework Based on QMIX

Building upon the state representation introduced above, we now describe the complete training workflow of the proposed HG-MARL strategy. At the beginning of each episode, the system constructs the heterogeneous graph $\mathcal{G}_t$ according to the physical positions and sensing ranges of all ES nodes. Each agent embeds its local observation and neighborhood information into the representation $\mathbf{h}_i^t$, and then uses this embedding to generate its individual action:

$$\mathbf{a}_i^t = \pi(\mathbf{h}_i^t) = (\mathbf{m}_i^t, \boldsymbol{\delta}_i^t), \quad (35)$$

where m_i^t denotes the trajectory control action and $\boldsymbol{\delta}_i^t$ represents the scheduling decision for selecting a subset of UAVs. Executing the joint action $\mathbf{a}^t = \{\mathbf{a}_i^t\}_{i=1}^{N_{ES}}$ yields the system-level reward R_t and the next state.

To enable coordination, each ES is equipped with an individual Q-network, and their outputs are integrated by a mixing network. The individual Q-function is defined as:

$$Q_i(\mathbf{h}_i^t, \mathbf{a}_i^t) = f_i(\mathbf{h}_i^t, \mathbf{a}_i^t; \theta_i), \quad (36)$$

which evaluates the value of a local action given the current state embedding. The global Q-value is then obtained by:

$$Q_{\text{tot}}(s^t, \mathbf{a}^t) = Mix_\omega(Q_1, Q_2, \ldots, Q_{N_{ES}}), \quad (37)$$

where ω are the parameters of the mixing network and s^t is the concatenated global state. In QMIX, the mixing network is constrained to satisfy

$$\frac{\partial Q_{\text{tot}}}{\partial Q_i} \geq 0, \quad \forall i, \quad (38)$$

so that improving individual policies cannot degrade the overall performance. Concretely, we follow the standard QMIX design where a hypernetwork generates the mixing weights and biases conditioned on s^t, and we enforce nonnegative mixing weights by applying an element-wise Softplus to the generated weights. This ensures $\partial Q_{\text{tot}}/\partial Q_i \geq 0$ by construction. The mixer parameters are generated by a state-dependent hypernetwork, which enhances adaptability to environmental dynamics. The temporal difference (TD) target of the global Q-value is

$$y_t = R_t + \gamma \cdot Q_{\text{tot}}^{-}(\mathbf{s}^{t+1}, \mathbf{a}^{t+1}), \quad (39)$$

where Q_{tot}^{-} is the target network estimation, and $\mathbf{a}^{t+1}$ is chosen greedily under the target networks. The corresponding loss function is

$$\mathcal{L} = \left(y_t - Q_{\text{tot}}(s^t, \mathbf{a}^t) \right)^2 . \quad (40)$$

The parameters of ϕ_θ, $\{Q_i\}$, and Mix_ω are updated jointly via backpropagation with mini-batch sampling, enabling end-to-end training of the graph-based encoder and the multi-agent policy networks.

During joint training, to improve consistency among distributed agents in dynamic environments, a lightweight neighbor-based parameter synchronization mechanism is employed as an alternative to centralized aggregation. Concretely, every N_{comm} steps, each agent synchronizes its parameters with neighboring agents through cooperative edges in the graph. This procedure involves sharing GNN layer weights and softly aggregating policy parameters, which mitigates policy divergence and enhances the consistency and stability of joint Q-value estimation. In our implementation, neighbor synchronization averages the trainable parameters of the GNN encoder ϕ_θ and the individual Q-networks $\{Q_i\}$, while the target networks are kept local and updated in the standard way. Each exchange sends a parameter vector whose size is proportional to the number of synchronized weights, and we do not use compression.

In the inference phase, each agent operates in a fully decentralized manner. Each ES builds a local graph $\mathcal{G}_t$ from its own observation and currently reachable neighbors (within sensing/communication range), without exchanging any global topology. The GNN encoder (with attention) maps the local graph to the state embedding $\mathbf{h}_i^t$, based on which the policy network selects the action $\mathbf{a}_i^t$. When link conditions change or a UAV/ES becomes unavailable, the local graph is updated accordingly; unavailable nodes/links are removed, and model uploading can switch to another reachable ES or buffer intermediate updates until connectivity is restored. During training, the individual Q-values are further combined by the QMIX mixer under the CTDE paradigm for coordination. This locality keeps both computation and information exchange tied

Algorithm 1 Joint Optimization Algorithm Based on HG-MARL

Input: Maximum training rounds N_{ep}, time step T, learning rates η_q, η_ϕ, discount factor γ, GNN encoder ϕ_θ, individual Q-networks $\{Q_i\}$, mixing network Mix_ω.
Initialization: initialize all network parameters including target copies ϕ_θ^-, Q_i^-, Mix_ω^-; initialize an empty replay buffer $\mathcal{R}$.
for $e = 1$ **to** N_{ep} **do**
  Reset environment and initialize graph $\mathcal{G}_t$ with observations $\mathbf{o}_i^t$.
  for $t = 1$ **to** T **do**
    Construct graph $\mathcal{G}_t$ based on ES positions and connectivity.
    Obtain embeddings: $\mathbf{h}_i^t = \phi_\theta(\mathbf{o}_i^t, \mathcal{G}_t)$.
    Select actions with ϵ-greedy: $\mathbf{a}_i^t = \pi(\mathbf{h}_i^t)$.
    Execute joint action $\mathbf{a}^t$, receive reward R_t, and next state $\mathbf{o}_i^{t+1}$.
    Store $(\mathcal{G}_t, \mathbf{o}_i^t, \mathbf{a}_i^t, R_t, \mathcal{G}_{t+1}, \mathbf{o}_i^{t+1})$ into $\mathcal{R}$.
  end for
  for each training step **do**
    Sample mini-batch $(\mathbf{s}, \mathbf{a}, R, \mathbf{s}')$ from $\mathcal{R}$.
    Recompute embeddings $\mathbf{h}_i^t = \phi_\theta(\mathbf{o}_i^t, \mathcal{G}_t)$ for the sampled $\mathbf{s}$.
    Compute individual Q-values: $Q_i(\mathbf{h}_i^t, \mathbf{a}_i^t)$.
    Compute global Q-value: $Q_{\text{tot}} = Mix_\omega(\{Q_i\})$.
    Compute target: y_t (with $\mathbf{a}^{t+1}$ chosen greedily under target networks).
    Minimize loss: $\mathcal{L}$.
    Update $\phi_\theta, \{Q_i\}, Mix_\omega$ via backpropagation.
    Periodically update target networks.
  end for
  Every N_{comm} steps, synchronize parameters among neighboring ES nodes.
end for
Output: trained ϕ_θ, $\{Q_i\}$, Mix_ω.

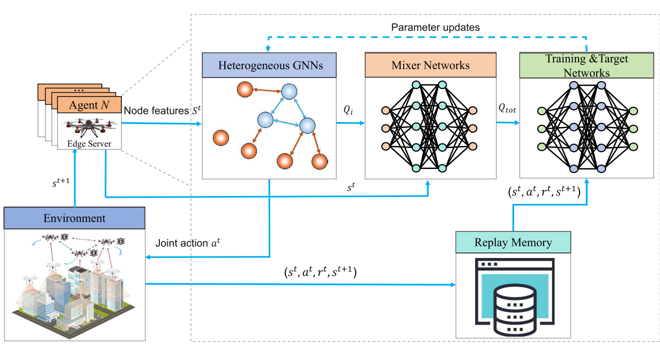


Fig. 4. HG-MARL policy training framework.

to the local neighborhood, which supports scalable deployment. Algorithm 1 details the joint optimization procedure, and Fig. 4 illustrates the overall closed-loop workflow.

To further clarify the overhead of the proposed framework, we discuss its time, memory, and communication complexity. Let $N = N_{\text{UAV}} + N_{\text{ES}}$ and $|E|$ denote the numbers of nodes and edges in $\mathcal{G}_t$, respectively. With an L-layer GNN encoder of hidden dimension d, the per-step graph encoding cost scales as $O(L|E|d)$, which is linear in the number of active links. The per-step value evaluation consists of (i) individual Q_i computation via a lightweight RNN and (ii) joint action-value aggregation via the QMIX mixing network. Up to constants, its cost scales as $O\big(N_a(d_q^2 + d_q A) + d_q(N_a + S)\big)$, where N_a is the number of learning agents, A is the action dimension, S is the global-state dimension, and d_q is the hidden size of the Q-networks/mixer. The main memory cost comes from the replay buffer and network parameters, dominated by storing transitions and thus on the order of $O(|\mathcal{D}|)$. In execution, each agent only relies on locally available neighbor information to build $\mathcal{G}_t$, so the required per-step communication/measurement overhead scales with the number of active local links.

VII. Simulation Results

To evaluate the coordination efficiency and decision-making capability of the layered UAVs-enabled ICCIC network in online DT channel modeling, we construct a 3D urban communication scenario for simulating channel measurement and data offloading tasks. Policy training and performance testing are conducted in a round-based manner. It should be emphasized that this study does not address the internal modeling of the DT itself, but focuses on layered UAVs coordination in data acquisition and offloading. The objective is to jointly optimize UAV trajectories and scheduling strategies to enable low-latency data offloading with minimal AoI, thereby providing high-quality and schedulable inputs for subsequent channel model construction.

A. Experimental Setup

The simulation platform in this section is built upon and extended from the open-source urban scene framework provided in [56]. While retaining its 3D city modeling structure and part of the original communication parameter settings, we modify the platform to support the layered UAVs architecture. The simulated environment covers an 800×600 m urban area populated with buildings and occlusion structures of varying heights, emulating a typical air-to-ground communication scenario with both LoS and NLoS channel conditions. The system randomly deploys 12 hovering working UAVs at altitudes of 30, 35, and 40 m for channel measurement, and 3 ESs at 70, 75, and 80 m for trajectory planning and data offloading reception. Each ES, as an independent agent, makes scheduling decisions based on GNN-encoded states. During each task episode, the ESs must return to their initial take-off positions to mark the end of the task. Training is conducted under the CTDE framework, where GNN-based state modeling is combined with QMIX optimization to minimize average service delay and AoI. Detailed system parameter configurations are summarized in Table I. In our QMIX-based training, we use the standard mixing network with hypernetworks and enforce nonnegative mixing weights via Softplus; during training we monitor the generated mixing weights as a sanity check for monotonicity.

To comprehensively evaluate the scheduling optimization performance of the proposed HG-MARL strategy within the

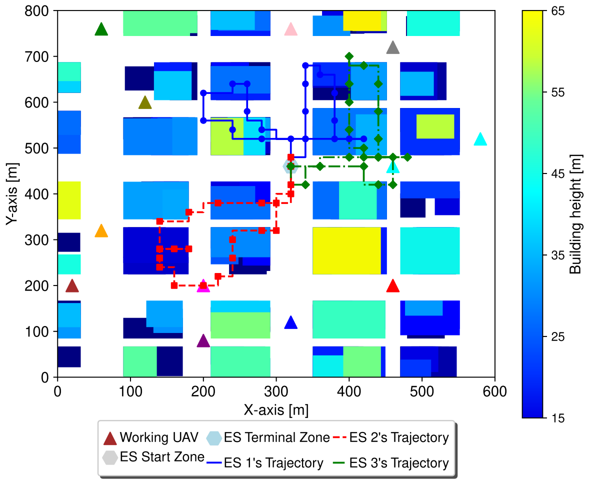


Fig. 5. Trajectory planning by HG-MARL in urban environment.

network, we focus on two key metrics. The first is the average service delay (Eq. (15)), reflecting the overall communication efficiency of ES scheduling during data offloading. The second is the average AoI (Eq. (10)), which quantifies the timeliness of the channel measurement data when it is uploaded for model construction.

To verify the effectiveness of the proposed method, we compare the following algorithm settings:

MAPPO: MAPPO is an actor–critic baseline with centralized training and decentralized execution, where a centralized critic is used during training to guide decentralized policies.

QPLEX: QPLEX is a value-decomposition baseline for multi-agent credit assignment, which models richer agent interactions than standard monotonic mixing.

Standard QMIX: The QMIX baseline constructs state inputs through conventional vector concatenation, without exploiting the topological characteristics of the communication graph.

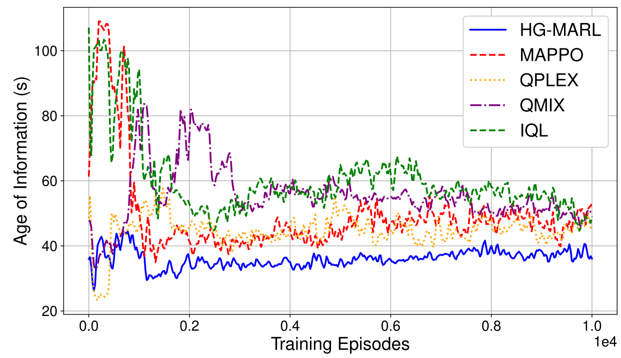
Independent Q-Learning (IQL): IQL treats each ES as an isolated agent, learning its policy independently without information sharing or task-level coordination.

All algorithms run under the same environment and parameter configuration to ensure fair and reproducible comparisons.

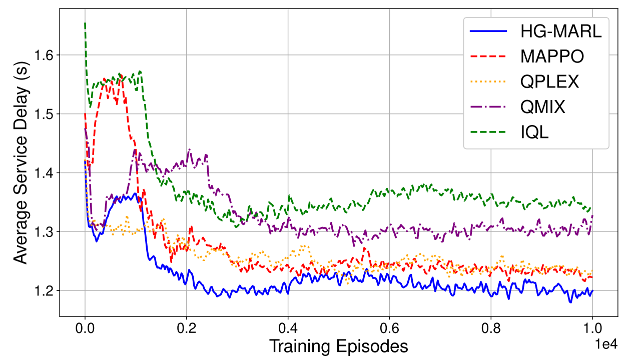
B. Experimental Results and Analysis

Fig. 5 illustrates representative trajectories of three ESs during a task episode in the urban environment. Each colored polyline shows the path of one ES, overlaid with the building layout (with height indicated by the color map) and the working UAV locations. The ESs tend to move along LoS-favorable corridors and avoid heavily blocked areas, which helps maintain more reliable links for data offloading and mitigates potential link congestion.

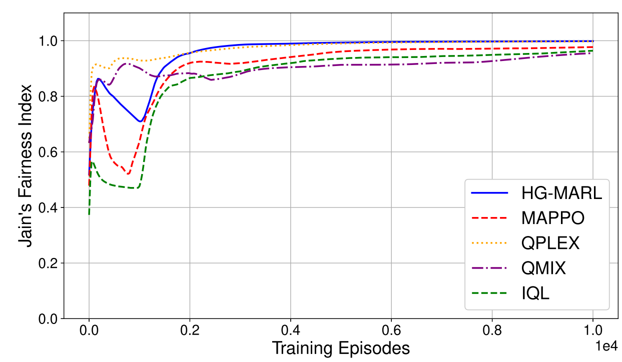
Moreover, the ESs spread over different regions and provide service in a spatially decoupled manner, which supports a more balanced coverage of working UAVs. These observations suggest that, with the heterogeneous graph modeling and the GNN-based feature aggregation, the learned policy can leverage environmental and link-related information to adapt its trajectory decisions in a complex urban scenario.



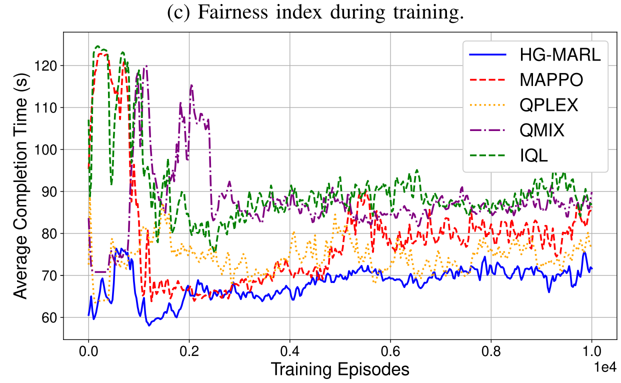
(a) Average AoI during training.



(b) Average service delay during training.



(c) Fairness index during training.



(d) Task completion time during training.

Fig. 6. Training performance comparison of different algorithms.

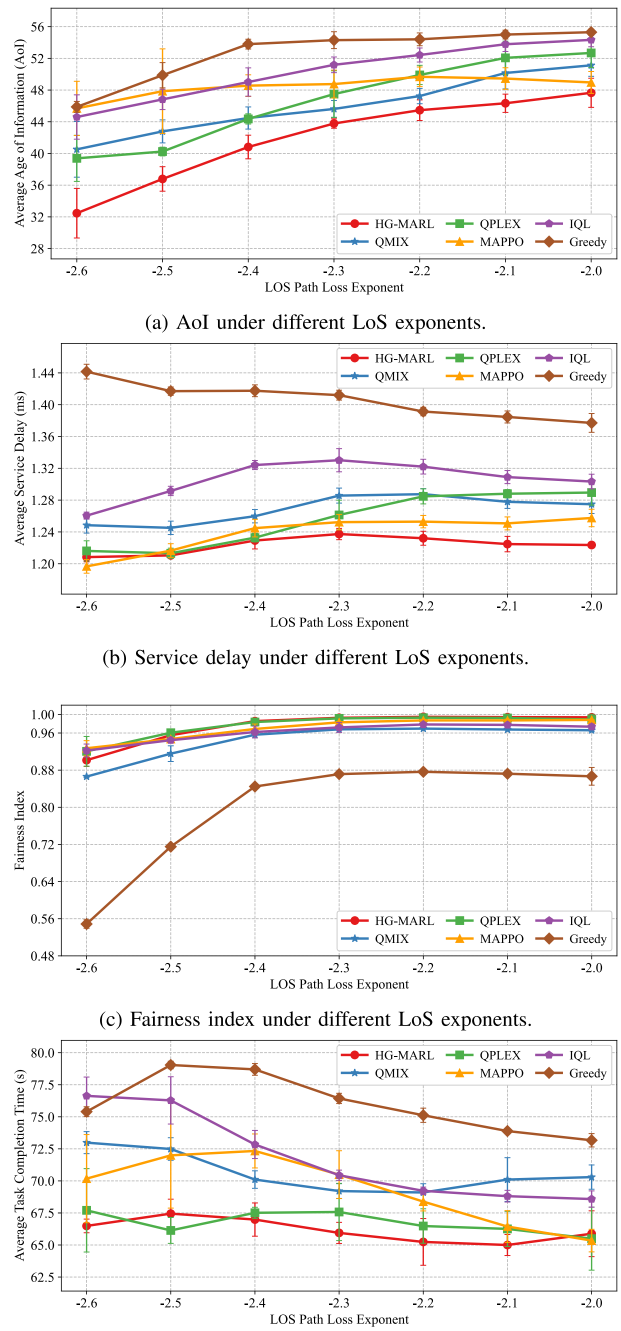
Fig. 6 illustrates the training behaviors of the compared algorithms on four metrics: (a) average AoI, (b) average service delay, (c) Jain's fairness index, and (d) task completion time. Overall, HG-MARL converges to a lower and more stable operating level after the initial exploration stage. Compared with baselines that rely on plain vector inputs (QMIX) or fully independent learning (IQL), HG-MARL reduces AoI more consistently, suggesting that the learned policy can better identify UAVs with stale updates and schedule service accordingly under time-varying links. A similar trend is observed for the service delay, where HG-MARL reaches a lower-delay region and remains stable, indicating improved coordination in offloading and resource allocation decisions.

Fairness is evaluated by Jain's index $\boldsymbol{J}\left(\overline{R}_t\right)$ and is guided by a soft penalty term with threshold η in the reward design, which helps mitigate long-term service bias while still focusing on timeliness. Finally, the task completion time reflects the overall execution efficiency of the learned scheduling strategy, and HG-MARL shows a faster stabilization behavior than the baseline methods. For visual clarity, Fig. 6 shows representative training curves, while the following comparisons report mean $\pm$ standard deviation over five independent runs.

Following the demonstration of training stability, we evaluate the learned policy under different channel conditions by varying the LoS path loss exponent. Fig. 7 compares HG-MARL with the baselines in terms of (a) average AoI, (b) average service delay, (c) task completion time, and (d) fairness index, with mean $\pm$ standard deviation over five independent runs. As the LoS exponent increases, all methods experience a clear performance degradation in AoI and delay due to the weaker link quality. Despite this change, HG-MARL achieves consistently low AoI and service delay across the full range, indicating that the learned scheduling decisions remain effective when link attenuation becomes more severe. Meanwhile, the fairness index of HG-MARL stays close to the target level, suggesting that timeliness is achieved without introducing evident long-term service bias. The completion-time curves show a consistent advantage of HG-MARL, reflecting more efficient execution across different LoS conditions.

We evaluate robustness under time-varying network conditions by introducing stochastic bandwidth and latency disturbances during execution. Specifically, the available bandwidth in each time slot is perturbed as $W(t) = W_0\left(1 + \varepsilon_t\right)$, where ε_t is a zero-mean random factor and its intensity is controlled by $\lambda \in [0, 1]$ (we consider both Gaussian fluctuations and piecewise jump changes). Here, λ scales the disturbance amplitude. In addition, we inject a delay jitter term $L(t)$ into the service latency to emulate random queuing/processing variations. We evaluate three disturbance modes: bandwidth variation only, delay jitter only, and their combined disturbance. The policy is trained under the default setting and is directly tested under disturbed conditions without retraining.

Fig. 8 reports the average AoI and average service delay versus λ, with mean $\pm$ standard deviation over five independent runs. Overall, the performance remains stable under single-source disturbances across the full range of λ. Under the combined disturbance, the variation becomes larger, but the policy does not exhibit unstable behavior. These results suggest that the learned scheduling strategy can adapt its decisions when link capacity and latency vary over time.



(a) AoI under different LoS exponents.

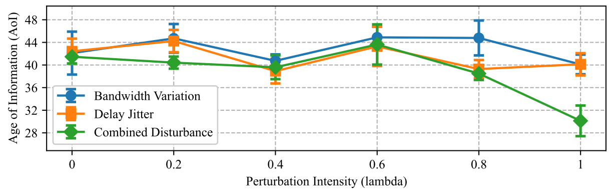
(b) Service delay under different LoS exponents.

(c) Fairness index under different LoS exponents.

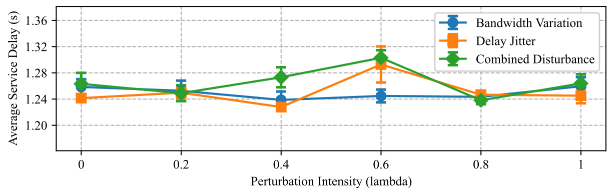
(d) Task completion time under different LoS exponents.

Fig. 7. Performance comparison under different LoS path loss exponents.

We also examine whether the objective form affects the learned scheduling behavior. The product form penalizes time slots where the system is both stale (high AoI) and slow (high delay), and it avoids a fixed tradeoff that can be overly sensitive to weight choices. In particular, we compare the proposed product-form objective with weighted-sum objectives under three representative weight configurations. All settings use the same training pipeline and evaluation protocol, and we report mean $\pm$ standard deviation over five independent runs. Table II summarizes the resulting average AoI and average service delay, showing how the product form leads to a more balanced timeliness outcome than the weighted sums.



(a) Average AoI versus perturbation intensity λ.



(b) Average service delay versus perturbation intensity λ.

Fig. 8. Robustness under bandwidth/latency disturbances (mean $\pm$ std).

TABLE II
OBJECTIVE-FORM COMPARISON (HG-MARL, MEAN $\pm$ STD, 5 SEEDS)

Objective	**Avg. AoI** (s)	**Avg. Delay** (s)
product	39.114 $\pm$ 1.381	1.2024 $\pm$ 0.0074
$w = 0.2/0.8$	41.016 $\pm$ 1.891	1.2867 $\pm$ 0.0134
$w = 0.5/0.5$	44.061 $\pm$ 2.816	1.2278 $\pm$ 0.0049
$w = 0.8/0.2$	45.314 $\pm$ 1.356	1.2447 $\pm$ 0.0102

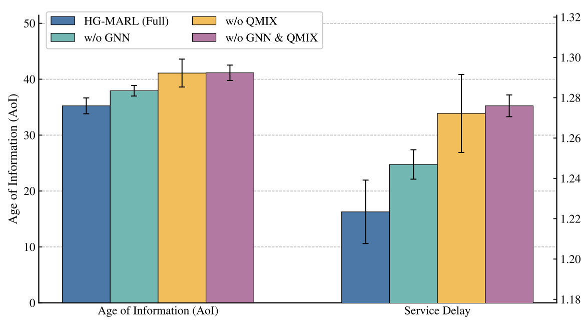
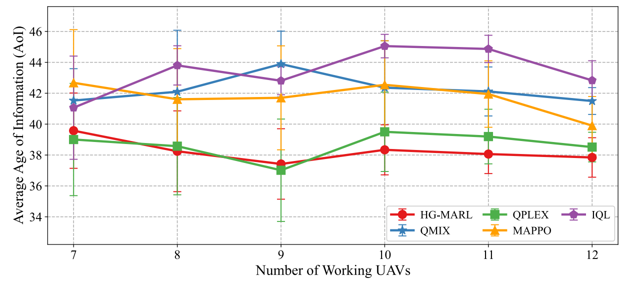


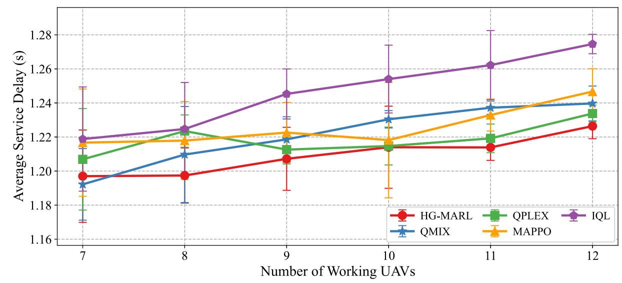
Fig. 9. Ablation study of HG-MARL (mean $\pm$ std).

To further understand which design modules contribute to the final performance, we conduct an ablation study on two key components of HG-MARL: the heterogeneous graph encoder (GNN) and the QMIX-based coordination module. We consider four variants: HG-MARL (Full), w/o GNN (replacing the graph encoder with a standard vector encoding), w/o QMIX (removing mixing-based coordination and learning independently), and w/o GNN & QMIX (removing both). Fig. 9 reports the resulting AoI and service delay with mean $\pm$ std over multiple runs. Removing either component leads to a clear performance drop, and the degradation becomes more pronounced when both are removed. This indicates that the GNN encoder and the QMIX coordination mechanism play complementary roles in improving timeliness while reducing the service delay.

We evaluate scalability by varying the system size through the number of working UAVs. Increasing the number of working UAVs raises the service demand and makes the scheduling problem more challenging. Fig. 10 reports the average AoI and average service delay under different numbers of working UAVs, with mean $\pm$ std over multiple runs. As the system size increases, all methods show a gradual performance degradation, which is expected due to higher contention for limited sub-channels and ES resources. Despite this change, HG-MARL maintains a consistent advantage over the baselines across all tested scales, indicating that the learned policy generalizes well to different system sizes and does not show unstable behavior.



(a) Average AoI versus the number of working UAVs.



(b) Average service delay versus the number of working UAVs.

Fig. 10. Performance under different numbers of working UAVs (mean $\pm$ std).

TABLE III
COMPLEXITY OF HG-MARL (PER STEP)

Component	**Time**	**Space**
GNN encoding	C_{gnn}	$O(Nd + \lvert E\rvert)$
Agent Q_i eval.	C_Q	$O(N_a d_q + \lvert\theta_{\text{rnn}}\rvert)$
QMIX mixing	C_{mix}	$O(\lvert\theta_{\text{mix}}\rvert + S d_q)$
Overall inference	$O(C_{\text{gnn}} + C_Q + C_{\text{mix}})$	$O(\lvert\theta_{\text{total}}\rvert + Nd + \lvert E\rvert)$
Replay buffer	–	$O(\lvert\mathcal{D}\rvert)$
Single train update	$O(BT(C_{\text{gnn}} + C_Q + C_{\text{mix}}))$	$O(BT \cdot \text{act.} + \lvert\theta_{\text{total}}\rvert)$
Communication	$O(\lvert E_{\text{local}}\rvert)$	$O(1)$

Note: $C_{\text{gnn}} = O(L\lvert E\rvert d)$, $C_Q = O(N_a(d_q^2 + d_q A))$, $C_{\text{mix}} = O(d_q(N_a + S))$.

We also report the computational overhead of HG-MARL to clarify its practical cost. Table III summarizes the per-step time and space complexity of the main modules, including the GNN encoder, the agent-level Q_i evaluation, and the QMIX mixing network, together with the overall inference cost and the replay-buffer storage. Table IV reports the measured runtime in our simulation setup, including model size, inference latency per step (and the corresponding throughput), and the time of one training update, reported as mean $\pm$ std over repeated runs. Since the decision period in our simulations is $\Delta t = 1$ s, the measured inference latency is below one time slot, which supports real-time scheduling under this setting.

TABLE IV
RUNTIME STATISTICS OF HG-MARL (MEAN ± STD OVER INDEPENDENT RUNS)

Algorithm	Params (M)	Inference (ms/step)	Throughput (steps/s)	Train update (ms)
HG-MARL	0.100	12.3964 ± 0.3297	80.7155 ± 2.2168	154.1820 ± 7.2310

From an implementation perspective, the learned policy can be executed in a decentralized manner at ESs using locally available inputs, including positions, link-quality indicators, and the AoI maintained from received updates. A full field validation with real UAVs and IoT devices requires system integration beyond the scope of this paper and will be considered in future work.

VIII. CONCLUSION

This paper tackles the challenge of achieving efficient and robust DT channel modeling in 6G networks by introducing a layered UAVs-enabled ICCIC network. In this architecture, mobile ESs coordinate with statically deployed working UAVs to enable continuous wireless channel measurement, data offloading, and edge computing in dynamic urban environments. We formulate a multi-objective joint scheduling problem for trajectory control and data offloading, using AoI and service delay as performance metrics under constraints on task completion time and service fairness. To capture the heterogeneous and time-varying interactions among UAVs, we construct a multi-relational heterogeneous graph and employ a GNN to extract structured state representations that encode both cooperative and service-level dependencies. Based on this representation, we design an HG-MARL framework that supports decentralized yet coordinated policy learning in partially observable settings. Simulation results demonstrate that the proposed method significantly reduces AoI and service delay, accelerates policy convergence, and maintains high robustness and adaptability in large-scale, dynamic urban scenarios, providing an effective pathway for real-time and reliable DT channel modeling.

REFERENCES

[1] G. Sun et al., "Resource management for integrated communications, computing, and sensing (ICCS) networks," *IEEE Trans. Veh. Technol.*, vol. 73, no. 12, pp. 18719–18731, Dec. 2024.

[2] Y. Gong, H. Yao, and A. Nallanathan, "Intelligent sensing, communication, computation, and caching for satellite-ground integrated networks," *IEEE Netw.*, vol. 38, no. 4, pp. 9–16, Jul. 2024.

[3] X. Lin, L. Kundu, C. Dick, E. Obiodu, T. Mostak, and M. Flaxman, "6G digital twin networks: From theory to practice," *IEEE Commun. Mag.*, vol. 61, no. 11, pp. 72–78, Nov. 2023.

[4] A. Masaracchia, V. Sharma, B. Canberk, O. A. Dobre, and T. Q. Duong, "Digital twin for 6G: Taxonomy, research challenges, and the road ahead," *IEEE Open J. Commun. Soc.*, vol. 3, pp. 2137–2150, 2022.

[5] C.-X. Wang et al., "An enhanced 6G pervasive channel model towards standardization," *Scientia Sinica Informationis*, vol. 55, no. 6, p. 1486, Jun. 2025.

[6] T. Liu et al., "6G integrated sensing and communications channel modeling: Challenges and opportunities," *IEEE Veh. Technol. Mag.*, vol. 19, no. 2, pp. 31–40, Jun. 2024.

[7] C.-X. Wang, Z. Lv, X. Gao, X. You, Y. Hao, and H. Haas, "Pervasive wireless channel modeling theory and applications to 6G GBSMs for all frequency bands and all scenarios," *IEEE Trans. Veh. Technol.*, vol. 71, no. 9, pp. 9159–9173, Sep. 2022.

[8] H. Jiang et al., "Deep reinforcement learning enabled UAV trajectory optimization for A2G communication systems," *IEEE Trans. Cognit. Commun. Netw.*, vol. 12, pp. 3164–3178, 2026.

[9] L. U. Khan, Z. Han, W. Saad, E. Hossain, M. Guizani, and C. S. Hong, "Digital twin of wireless systems: Overview, taxonomy, challenges, and opportunities," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 4, pp. 2230–2254, 4th Quart., 2022.

[10] Z. Feng, Z. Wei, X. Chen, H. Yang, Q. Zhang, and P. Zhang, "Joint communication, sensing, and computation enabled 6G intelligent machine system," *IEEE Netw.*, vol. 35, no. 6, pp. 34–42, Nov. 2021.

[11] P. Kumar, S. Bhattacharyya, S. Darshi, S. Majhi, A. A. Almohammedi, and S. Shailendra, "Outage analysis using probabilistic channel model for drone assisted multi-user coded cooperation system," *IEEE Trans. Veh. Technol.*, vol. 72, no. 8, pp. 10273–10285, Aug. 2023.

[12] W. Xu, T. Zhang, X. Mu, Y. Liu, and Y. Wang, "Trajectory planning and resource allocation for multi-UAV cooperative computation," *IEEE Trans. Commun.*, vol. 72, no. 7, pp. 4305–4318, Jul. 2024.

[13] S. Ren, J. Zhou, S. Gao, and C. C. Chen, "Cooperative fission control algorithm of UAV swarm based on attention mechanism," *Acta Electronica Sinica*, vol. 51, no. 7, pp. 1898–1905, 2023.

[14] H. Zhao et al., "Online trajectory optimization for energy-efficient cellular-connected UAVs with map reconstruction," *IEEE Trans. Veh. Technol.*, vol. 73, no. 3, pp. 3445–3456, Mar. 2024.

[15] D. Xu and G. Chen, "The research on intelligent cooperative combat of UAV cluster with multi-agent reinforcement learning," *Aerosp. Syst.*, vol. 5, no. 1, pp. 107–121, Mar. 2022.

[16] R. D. Yates et al., "Age of information: An introduction and survey," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 5, pp. 1183–1210, May 2021.

[17] Y. Emami, H. Gao, K. Li, L. Almeida, E. Tovar, and Z. Han, "Age of information minimization using multi-agent UAVs based on AI-enhanced mean field resource allocation," *IEEE Trans. Veh. Technol.*, vol. 73, no. 9, pp. 13368–13380, Sep. 2024.

[18] Y. Liao and V. Friderikos, "Energy and age Pareto optimal trajectories in UAV-assisted wireless data collection," *IEEE Trans. Veh. Technol.*, vol. 71, no. 8, pp. 9101–9106, Aug. 2022.

[19] M. Sheraz, T. C. Chuah, Y. L. Lee, M. M. Alam, A. Al-Habashna, and Z. Han, "A comprehensive survey on revolutionizing connectivity through artificial intelligence-enabled digital twin network in 6G," *IEEE Access*, vol. 12, pp. 49184–49215, 2024.

[20] C.-X. Wang et al., "Modeling, capacity studies, antenna and system designs for 6G/B6G 3-D continuous-space radio channels enabled by electromagnetic information theory," *IEEE Commun. Surveys Tuts.*, vol. 28, pp. 1–63, 2026.

[21] J. Li, C.-X. Wang, C. Huang, T. Qi, and T. Wu, "Digital twin online channel modeling: Challenges, principles, and applications," *IEEE Veh. Technol. Mag.*, vol. 20, no. 1, pp. 94–103, Mar. 2025.

[22] X. Gong et al., "Digital twin of channel: Diffusion model for sensing-assisted statistical channel state information generation," *IEEE Trans. Wireless Commun.*, vol. 24, no. 5, pp. 3805–3821, May 2025.

[23] H. Jiang et al., "Large-scale RIS enabled air-ground channels: Near-field modeling and analysis," *IEEE Trans. Wireless Commun.*, vol. 24, no. 2, pp. 1074–1088, Feb. 2025.

[24] B. Li, W. Xie, Y. Ye, L. Liu, and Z. Fei, "FlexEdge: Digital twin-enabled task offloading for UAV-aided vehicular edge computing," *IEEE Trans. Veh. Technol.*, vol. 72, no. 8, pp. 11086–11091, Aug. 2023.

[25] Y. Li, D. V. Huynh, T. Do-Duy, E. Garcia-Palacios, and T. Q. Duong, "Unmanned aerial vehicle-aided edge networks with ultra-reliable low-latency communications: A digital twin approach," *IET Signal Process.*, vol. 16, no. 8, pp. 897–908, Oct. 2022.

[26] B. Li, W. Liu, W. Xie, N. Zhang, and Y. Zhang, "Adaptive digital twin for UAV-assisted integrated sensing, communication, and computation networks," *IEEE Trans. Green Commun. Netw.*, vol. 7, no. 4, pp. 1996–2009, Dec. 2023.

[27] P. Consul, I. Budhiraja, D. Garg, N. Kumar, R. Singh, and A. S. Almogren, "A hybrid task offloading and resource allocation approach for digital twin-empowered UAV-assisted MEC network using federated reinforcement learning for future wireless network," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 3120–3130, Feb. 2024.

[28] H. Zhao, M. Sui, M. Liu, C. Zhu, and H. Zhu, "An incentive assignment scheme of UAV clients for federated intelligent inspection based on communication-sensing-computing integration," *IEEE Trans. Mobile Comput.*, early access, 2026, doi: 10.1109/TMC.2026.3651590.

[29] B. Li, Y. Liu, L. Tan, H. Pan, and Y. Zhang, "Digital twin assisted task offloading for aerial edge computing and networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 10, pp. 10863–10877, Oct. 2022.

[30] F. Qi, W. Xie, L. Liu, T. Hong, and F. Zhou, "UAV digital twin based wireless channel modeling for 6G green IoT," *Drones*, vol. 7, no. 9, p. 562, Sep. 2023.

[31] K. Liang et al., "Digital twin assisted deep reinforcement learning for computation offloading in UAV systems," *IEEE Trans. Veh. Technol.*, vol. 74, no. 5, pp. 8466–8471, May 2025.

[32] G. Khalaf, M. Itani, and S. Sharafeddine, "A UAV-aided digital twin framework for IoT networks with high accuracy and synchronization," 2025, *arXiv:2504.15967*.

[33] X. Wang, Z. Fei, Y. Zhou, and J. Hu, "Integrated sensing, communication, computation, and intelligence towards IoT: Key technologies and future directions," *J. Electron. Inf. Technol.*, vol. 47, no. 4, pp. 888–908, 2025.

[34] G. Chen, C. Cheng, X. Xu, and Y. Zeng, "Minimizing the age of information for data collection by cellular-connected UAV," *IEEE Trans. Veh. Technol.*, vol. 72, no. 7, pp. 9631–9635, Jul. 2023.

[35] O. S. Oubbati, J. Alotaibi, F. Alromithy, M. Atiquzzaman, and M. R. Altimania, "A UAV-UGV cooperative system: Patrolling and energy management for urban monitoring," *IEEE Trans. Veh. Technol.*, vol. 74, no. 9, pp. 13521–13536, Sep. 2025.

[36] J. Du, H. Wang, C. Jiang, J. Simonjan, J. Wang, and M. Debbah, "Distributed AI-based secure communications in space-air-ground-sea integrated networks," *IEEE Commun. Mag.*, vol. 63, no. 7, pp. 48–55, Jul. 2025.

[37] H. Zhang et al., "Joint optimization of transmit power and trajectory for UAV-enabled data collection with dynamic constraints," *IEEE Trans. Commun.*, vol. 73, no. 9, pp. 8080–8091, Sep. 2025.

[38] X. Yao et al., "Joint optimization of UAV trajectory and resource allocation for federated learning," *J. Comput. Eng. Appl.*, vol. 60, no. 11, pp. 336–345, 2024.

[39] J. Cui, Y. Liu, and A. Nallanathan, "Multi-agent reinforcement learning-based resource allocation for UAV networks," *IEEE Trans. Wireless Commun.*, vol. 19, no. 2, pp. 729–743, Feb. 2020.

[40] Q. Wu, Z. Geng, Y. Ren, Q. Feng, and J. Zhong, "Multi-UAV redeployment optimization based on multi-agent deep reinforcement learning oriented to swarm performance restoration," *Sensors*, vol. 23, no. 23, p. 9484, Nov. 2023.

[41] G. Sun et al., "Multi-objective optimization for multi-UAV-assisted mobile edge computing," *IEEE Trans. Mobile Comput.*, vol. 23, no. 12, pp. 14803–14820, Dec. 2024.

[42] Y. Zeng, X. Xu, S. Jin, and R. Zhang, "Simultaneous navigation and radio mapping for cellular-connected UAV with deep reinforcement learning," *IEEE Trans. Wireless Commun.*, vol. 20, no. 7, pp. 4205–4220, Jul. 2021.

[43] Y. He, Y. Gan, H. Cui, and M. Guizani, "Fairness-based 3-D multi-UAV trajectory optimization in multi-UAV-assisted MEC system," *IEEE Internet Things J.*, vol. 10, no. 13, pp. 11383–11395, Jul. 2023.

[44] S. Tan, B. Chen, D. Liu, J. Zhang, and L. Hanzo, "Communication-assisted multi-agent reinforcement learning improves task-offloading in UAV-aided edge-computing networks," *IEEE Wireless Commun. Lett.*, vol. 12, no. 12, pp. 2233–2237, Dec. 2023.

[45] O. S. Oubbati, N. Chaib, A. Lakas, and S. Bitam, "On-demand routing for urban VANETs using cooperating UAVs," in *Proc. Int. Conf. Smart Commun. Netw. Technol. (SaCoNeT)*, Oct. 2018, pp. 108–113.

[46] Y. Yang et al., "Channel knowledge map for cellular-connected UAV via binary Bayesian filtering," *IEEE Trans. Commun.*, vol. 73, no. 10, pp. 9222–9235, Oct. 2025.

[47] B. Zhao et al., "Graph-based multi-agent reinforcement learning for collaborative search and tracking of multiple UAVs," *Chin. J. Aeronaut.*, vol. 38, no. 3, Mar. 2025, Art. no. 103214.

[48] P. Li, L. Wang, W. Wu, F. Zhou, B. Wang, and Q. Wu, "Graph neural network-based scheduling for multi-UAV-enabled communications in D2D networks," *Digit. Commun. Netw.*, vol. 10, no. 1, pp. 45–52, Feb. 2024.

[49] X. Zhang, H. Zhao, J. Wei, C. Yan, J. Xiong, and X. Liu, "Cooperative trajectory design of multiple UAV base stations with heterogeneous graph neural networks," *IEEE Trans. Wireless Commun.*, vol. 22, no. 3, pp. 1495–1509, Mar. 2023.

[50] C.-X. Wang, Z. Lv, Y. Chen, and H. Haas, "A complete study of space-time-frequency statistical properties of the 6G pervasive channel model," *IEEE Trans. Commun.*, vol. 71, no. 12, pp. 7273–7287, Dec. 2023.

[51] J. Won, D.-Y. Kim, Y.-I. Park, and J.-W. Lee, "A survey on UAV placement and trajectory optimization in communication networks: From the perspective of air-to-ground channel models," *ICT Exp.*, vol. 9, no. 3, pp. 385–397, Jun. 2023.

[52] Y. Pan et al., "GNN-empowered effective partial observation MARL method for AoI management in multi-UAV network," *IEEE Internet Things J.*, vol. 11, no. 21, pp. 34541–34553, Nov. 2024.

[53] H. Jiang, W. Shi, X. Chen, Q. Zhu, and Z. Chen, "High-efficient near-field channel characteristics analysis for large-scale MIMO communication systems," *IEEE Internet Things J.*, vol. 12, no. 6, pp. 7446–7458, Mar. 2025.

[54] O. A. Amodu et al., "Deep reinforcement learning for AoI minimization in UAV-aided data collection for WSN and IoT: A survey," *IEEE Access*, vol. 12, pp. 108000–108040, 2024.

[55] J. Alotaibi, O. S. Oubbati, M. Atiquzzaman, F. Alromithy, and M. R. Altimania, "Optimizing disaster response with UAV-mounted RIS and HAP-enabled edge computing in 6G networks," *J. Netw. Comput. Appl.*, vol. 241, Sep. 2025, Art. no. 104213.

[56] J. Chen et al., "Model-aided federated reinforcement learning for multi-UAV trajectory planning in IoT networks," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, Jul. 2023, pp. 818–823.

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